

Assignment_CloudFormation_1

AWS CloudFormation: Template, EC2, VPC, and S3

1. What is CloudFormation? Explain briefly.

AWS CloudFormation is an Infrastructure as Code service from Amazon Web Services. It allows us to describe cloud resources in a template file, usually written in YAML or JSON. Instead of creating resources manually from the AWS Console, we define them in the template and CloudFormation creates them for us.

The template can include resources such as VPCs, EC2 instances, S3 buckets, security groups, databases, IAM roles, and many other AWS services. When the template is deployed, CloudFormation creates a stack. A stack is a group of AWS resources that are managed together as one unit.

CloudFormation is useful because it makes infrastructure repeatable, trackable, and easier to manage. If the same environment is needed again, the same template can be reused. If something needs to change, we update the template and CloudFormation applies the change in a controlled way.

Brief deployment flow

- 1 Write the required resources inside a CloudFormation template.
- 2 Create a stack from the template in AWS CloudFormation.
- 3 CloudFormation reads dependencies and creates resources in the correct order.
- 4 After deployment, the Outputs section shows useful values such as VPC ID, EC2 instance ID, public IP, and S3 bucket name.

2. CloudFormation Template (1/3)

```
AWSTemplateFormatVersion: '2010-09-09'
Description: Create a VPC, public EC2 instance, and S3 bucket.

Parameters:
  LatestAmiId:
    Type: AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>
    Default: /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64

Resources:
  DemoVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true
      Tags:
        - Key: Name
          Value: Demo-VPC

  PublicSubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref DemoVPC
      CidrBlock: 10.0.1.0/24
      MapPublicIpOnLaunch: true
      Tags:
        - Key: Name
          Value: Demo-Public-Subnet

  InternetGateway:
    Type: AWS::EC2::InternetGateway
    Properties:
      Tags:
        - Key: Name
          Value: Demo-IGW

  AttachGateway:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref DemoVPC
      InternetGatewayId: !Ref InternetGateway

  PublicRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref DemoVPC
      Tags:
        - Key: Name
          Value: Demo-Public-RouteTable

  DefaultRoute:
    Type: AWS::EC2::Route
    DependsOn: AttachGateway
    Properties:
      RouteTableId: !Ref PublicRouteTable
```

2. CloudFormation Template (2/3)

```
DestinationCidrBlock: 0.0.0.0/0
GatewayId: !Ref InternetGateway

SubnetRouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PublicSubnet
    RouteTableId: !Ref PublicRouteTable

InstanceSecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Allow HTTP and SSH access
    VpcId: !Ref DemoVPC
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 22
        ToPort: 22
        CidrIp: 0.0.0.0/0
      - IpProtocol: tcp
        FromPort: 80
        ToPort: 80
        CidrIp: 0.0.0.0/0
    Tags:
      - Key: Name
        Value: Demo-SG

DemoEC2Instance:
  Type: AWS::EC2::Instance
  Properties:
    ImageId: !Ref LatestAmiId
    InstanceType: t2.micro
    SubnetId: !Ref PublicSubnet
    SecurityGroupIds:
      - !Ref InstanceSecurityGroup
    Tags:
      - Key: Name
        Value: Demo-EC2
    UserData:
      Fn::Base64: !Sub |
        #!/bin/bash
        dnf install -y httpd
        systemctl enable httpd
        systemctl start httpd
        echo "CloudFormation EC2 Demo" > /var/www/html/index.html

DemoS3Bucket:
  Type: AWS::S3::Bucket
  Properties:
    PublicAccessBlockConfiguration:
      BlockPublicAcls: true
      BlockPublicPolicy: true
      IgnorePublicAcls: true
      RestrictPublicBuckets: true
    BucketEncryption:
```

2. CloudFormation Template (3/3)

```
    ServerSideEncryptionConfiguration:
      - ServerSideEncryptionByDefault:
          SSEAlgorithm: AES256

Outputs:
  VPCId:
    Description: Created VPC ID
    Value: !Ref DemoVPC

  EC2InstanceId:
    Description: Created EC2 instance ID
    Value: !Ref DemoEC2Instance

  EC2PublicIP:
    Description: Public IP address of the EC2 instance
    Value: !GetAtt DemoEC2Instance.PublicIp

  S3BucketName:
    Description: Created S3 bucket name
    Value: !Ref DemoS3Bucket
```

Sample Stack Output

After the stack is created successfully, CloudFormation shows useful output values from the Outputs section. The exact IDs and names will be different in every AWS account, but the output will look similar to this:

Output Key	Example Value	Meaning
VPCId	vpc-0a12b34c56d78e901	ID of the VPC created by the stack.
EC2InstanceId	i-0123abcd4567ef890	ID of the EC2 instance.
EC2PublicIP	13.201.10.25	Public IP used to access the web page.
S3BucketName	demo-stack-demos3bucket-a1b2c3d4	Name of the encrypted S3 bucket.

Note

The template opens SSH and HTTP to 0.0.0.0/0 for learning purposes. In a real project, SSH access should be restricted to a trusted IP address.